INDIAN INSTITUTE OF INFORMATION TECHNOLOGY KALYANI

Department of Computer Science and Engineering

Software Engineering Project Assignment

MediFlow

Hospital Management System

Assignment 1 · Hospital Management System

Software Requirements Specification (SRS)

GROUP 11

Sl.	Name of the member	Registration number
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Live demo at mediflow-hms-6vw1.onrender.com

Source code at github.com/anipriyo/mediflow-hms

Submitted through Google Classroom · October 2026

Group Details

Table 1. Group and assignment details

Field	Value
Group number	11
Assignment number and title	Assignment 1, Hospital Management System
Document	Software Requirements Specification (SRS) (document 1 of 5)
Project name	MediFlow Hospital Management System
Problem statement (brief)	Develop a system to manage patients, doctors, appointments, prescriptions, billing and medical records.
Companion documents	Software Design Document (SDD), UML Diagrams, Software Test Plan, Software Test Report
Live demo	mediflow-hms-6vw1.onrender.com
Source code	github.com/anipriyo/mediflow-hms

Table 2. Group members (names and registration numbers)

Sl.	Name	Registration number
1	Akash Chakraborty	CSE/23011/1059
2	Anipriyo Mandal	CSE/23017/1065
3	Arpan Mandal	CSE/23027/1075
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Document control

Table 3. Revision history

Version	Date	Description
0.1	15-Oct-2026	Project selected (Assignment 1, Hospital Management System)
0.2	18-Oct-2026	Scope, user roles and feature set agreed (core, hospital operations, distinctive features)
0.5	21-Oct-2026	Requirements baselined (50 FR, 10 NFR, 12 BR), architecture and UML diagrams
0.9	21-Oct-2026	Prototype implemented, unit, integration, system and acceptance tests executed
0.95	22-Oct-2026	Robustness / edge-case testing, additional defects fixed and regression-tested
1.0	22-Oct-2026	Final report for submission
1.1	22-Oct-2026	Prototype deployed online (Render), live demo link added
1.2	25-Oct-2026	Issued as a separate document (Software Requirements Specification (SRS)), one of five

Summary

MediFlow is a web-based Hospital Management System that manages **patients, doctors, appointments, prescriptions, billing and medical records**, extended with laboratory, pharmacy and in-patient (bed and admission) management. Beyond the core requirements, it introduces a **smart queue** with tokens and waiting-time estimates, a **live bed board** with a discharge workflow tied to billing clearance, and **role-based access control with an append-only audit trail** for six staff roles.

This document specifies 50 functional and 10 non-functional requirements, 12 business rules and 14 user requirements, together with the problem statement, objectives, scope, system requirements, hardware and software requirements, assumptions and constraints. It is the baseline for the other four documents.

> **Live demo.** The prototype runs online at mediflow-hms-6vw1.onrender.com and opens in any browser with no installation. Sign in with one of the demo accounts (admin, reception, dr.sharma, lab, pharmacy and billing, all with the password Mediflow@123). It uses a free cloud plan, so the first visit after 15 idle minutes takes about a minute to load, and the demo data is reset to fresh records dated that day whenever the server restarts.
>
> **Source code and running it locally.** The full source code is at github.com/anipriyo/mediflow-hms. Install Node.js 20 or later, run `npm install` and then `npm run dev` in the project folder, and open localhost port 5173 in a browser.

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SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

> This SRS follows the structure required by the assignment and is informed by IEEE Std 830-1998 / ISO/IEC/IEEE 29148 (2018). Every requirement has a unique identifier (FR-xx, NFR-xx, BR-xx, UR-xx) that is traced to design elements and test cases in the Software Test Report and its traceability matrix (Appendix A).

1. Introduction

1.1 Purpose

This document specifies the requirements of **MediFlow**, a web-based Hospital Management System (HMS) developed for Assignment 1 of the Software Engineering course. It describes what the system must do (functional requirements), how well it must do it (non-functional requirements), who will use it and under which assumptions and constraints. It is the baseline for the Software Design Document, the UML diagrams, the Software Test Plan and the Software Test Report.

1.2 Intended audience

- **Course instructor and evaluators**, to assess completeness and quality of the specification.
- **Development team (Group 11)**, as the single source of truth for design, implementation and testing.
- **Hospital stakeholders (future)**, administrators, doctors, front-desk, laboratory, pharmacy and billing staff who would validate the requirements.

1.3 Product overview

MediFlow connects the reception desk, doctors, laboratory, pharmacy, wards and billing through one shared patient record. A patient's journey (**registration → appointment → check-in and queue token → consultation → prescription / lab tests → billing → payment**, and for in-patients **admission → bed allocation → transfer → discharge**) flows through the system without re-entering data. Three features distinguish the product from a basic HMS. These are a **smart queue** with tokens and waiting-time estimates, a **live bed board** with a safe admission/discharge workflow tied to billing clearance, and **role-based access control with an append-only audit trail**.

1.4 Definitions, acronyms and abbreviations

Table 4. Definitions and acronyms

Term	Meaning
HMS	Hospital Management System
OPD / IPD	Out-patient department (visits without admission) / In-patient department (admitted patients)
RBAC	Role-Based Access Control, permissions are granted to roles, users get one role
Token	Sequential queue number issued to a patient at check-in, unique per doctor per day
ETA	Estimated waiting time until the patient is called
STAT	Urgent laboratory priority (processed before ROUTINE orders)
Charge	A billable event (consultation fee, lab test, medicine, bed-day) waiting to be invoiced
Billing clearance	Condition that all charges of an admission are invoiced and fully paid
JWT	JSON Web Token, signed session token used to authenticate API calls
SPA / REST	Single-Page Application / Representational State Transfer (HTTP + JSON API)
FR / NFR / BR / UR	Functional / Non-functional requirement, Business rule, User requirement

1.5 References

- IEEE Std 830-1998, *Recommended Practice for Software Requirements Specifications*. ISO/IEC/IEEE 29148 (2018), *Requirements engineering*.
- IEEE Std 1016-2009, *Software Design Descriptions*. IEEE Std 829-2008 / ISO/IEC/IEEE 29119-3, *Software test documentation*.
- OMG Unified Modeling Language (UML) 2.5.1 specification.
- Assignment brief. “Software Development project problems, 1. Hospital Management System”, Software Engineering course, 2026.

1.6 Document overview

Section 2 states the problem, Section 3 the objectives and Section 4 the scope. Sections 5 and 6 list functional and non-functional requirements, Section 7 the user requirements, Section 8 the detailed system requirements (features, interfaces and data), Section 9 the hardware and software requirements, and Section 10 the assumptions and constraints.

2. Problem Statement

Many small and mid-sized hospitals still run their front desk, wards and billing on paper registers and disconnected spreadsheets. This causes recurring problems.

- **Duplicate and incomplete patient records**, the same patient is registered several times, so history, allergies and previous results are not available to the doctor.
- **Double-booked doctors and long, unpredictable waits**, appointments are written in diaries, walk-ins and booked patients compete, and nobody can tell a patient how long they will wait.
- **Lost or illegible clinical information**, prescriptions and lab orders on paper are misread, results arrive late and abnormal values are not highlighted.
- **Bed confusion**, the front desk does not know which beds are free, being cleaned or under maintenance. Occasionally a bed is promised to two patients.
- **Revenue leakage**, consultation fees, tests, medicines and bed-days are recorded in different places, invoices are compiled by hand, and patients are sometimes discharged with unpaid bills.
- **No accountability**, it is impossible to tell who changed a record, cancelled an appointment or authorised a discharge.

The hospital therefore needs a single, secure system that manages **patients, doctors, appointments, prescriptions, billing and medical records** (as required by the assignment) as one continuous workflow, with the right information available to the right staff member at the right time.

3. Objectives

Table 5. Project objectives

ID	Objective	Measure of success
O1	Maintain one permanent electronic record per patient with complete visit, clinical, laboratory, admission and billing history.	Unique patient ID, duplicate warning (FR-07 to FR-12).
O2	Eliminate double-booking and make waiting times predictable.	Zero overlapping bookings, token + ETA for every checked-in patient (FR-15 to FR-25).
O3	Digitise consultations, prescriptions and lab orders and flag abnormal results automatically.	Diagnosis mandatory, abnormal flag on out-of-range results (FR-26 to FR-34).

ID	Objective	Measure of success
O4	Control medicine stock and prevent dispensing of expired or unavailable drugs.	Dispensing blocked when short/expired, alerts for low/expiring stock (FR-35 to FR-37).
O5	Give real-time bed visibility and a safe admission → transfer → discharge workflow.	No bed double-allocation, discharge only after billing clearance (FR-38 to FR-43).
O6	Capture every billable event automatically and produce accurate invoices and receipts.	Invoices built from charges, overpayment refused (FR-44 to FR-47).
O7	Protect patient data and make every action traceable.	RBAC on every API, lock-out, append-only audit log (FR-01 to FR-06, FR-50).
O8	Provide managers with live KPIs and period reports.	Role dashboards and admin reports (FR-48, FR-49).

4. Scope of the System

4.1 In scope

- **Patient management**, registration with unique ID, validation, duplicate detection, search, update and full history.
- **Doctor and department management**, profiles, fees, working days/hours, slot length and average consultation time.
- **Appointments**, free-slot calculation, booking, rescheduling, cancellation and no-show handling.
- **Smart queue**, check-in tokens, live queue board, waiting-time estimates, urgent prioritisation.
- **Consultation and medical records**, vitals, symptoms, diagnosis, notes, follow-up, e-prescriptions, lab orders.
- **Laboratory**, worklist, sample collection, result entry and automatic abnormal flagging.
- **Pharmacy**, inventory with batch/expiry, restocking, low-stock and expiry alerts, dispensing.
- **In-patient management**, wards and beds, bed board, admission, transfer, discharge request and discharge with billing clearance.
- **Billing**, automatic charges, invoices, discounts, part payments (cash, card, UPI), receipts and printing.
- **Administration**, users and roles, dashboards, management reports and audit trail.

4.2 Out of scope (this release)

- Patient self-service portal and online appointment booking by patients.
- Online payment gateway and insurance / TPA claim processing (payments are recorded, not processed).
- Integration with laboratory analysers, PACS / radiology imaging and external pharmacies.
- SMS / e-mail notifications, staff rostering and payroll, operation-theatre scheduling, inventory purchasing.
- AI-based features (e.g. clinical note summarisation), excluded for privacy and reliability reasons.

4.3 User classes

Table 6. User classes (actors)

Actor	Description and main responsibilities
Administrator	Manages users, doctors, departments, wards/beds and system configuration, views reports and the audit trail. May approve discharge overrides.
Receptionist	Registers patients, books/reschedules/cancels appointments, checks patients in (queue tokens), admits patients, allocates/transfers beds and completes discharge.
Doctor	Works through own queue, records consultations, writes prescriptions, orders lab tests, views patient history and lab results, requests admission/discharge.

Actor	Description and main responsibilities
Lab Technician	Processes lab test orders. Records sample collection and enters results.
Pharmacist	Maintains medicine inventory, receives stock alerts and dispenses prescriptions.
Billing Staff	Generates invoices from unbilled charges, applies discounts, records payments and prints receipts. Clears patients for discharge.

Patients are **external** to the system. They do not log in. Their data is entered and used by staff.

Use Case Diagram 1 - System overview (context level)

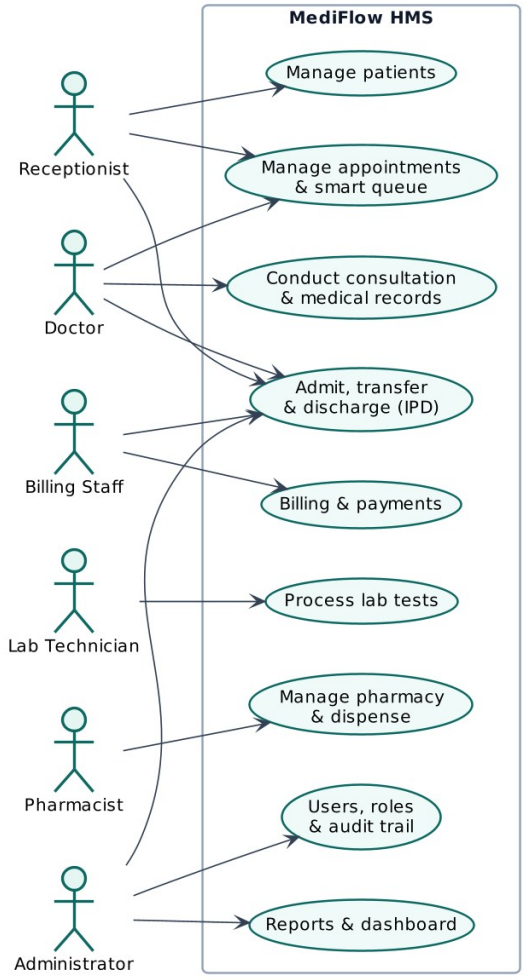


Figure 1. System context (actors and major functions)

5. Functional Requirements

The system shall satisfy the 50 functional requirements below, grouped into 13 modules. Priority follows MoSCoW-style ranking. **High** = must have for the first release, **Medium** = should have, **Low** = could have. All requirements were implemented in the prototype.

5.1 Business rules

Table 7. Business rules referenced by the functional requirements

ID	Business rule
BR-01	Patient ID format is P followed by 5 digits, assigned sequentially and never reused.
BR-02	Appointment slots start at the doctor's work start time and repeat every slot length until work end. Bookings must align to a slot boundary.
BR-03	Active appointment states are SCHEDULED, CHECKED_IN and IN_CONSULTATION, only these block a slot.

ID	Business rule
BR-04	Appointment life cycle. SCHEDULED → CHECKED_IN → IN_CONSULTATION → COMPLETED. SCHEDULED → CANCELLED. SCHEDULED → NO_SHOW.
BR-05	Queue order. URGENT before NORMAL, then earlier check-in first. Wait estimate = ahead × average consultation minutes + max(average − elapsed, 1) when a consultation is in progress.
BR-06	Lab order life cycle. ORDERED → SAMPLE_COLLECTED → COMPLETED. ORDERED → CANCELLED.
BR-07	Low stock when quantity ≤ reorder level. Near-expiry when expiry is within 30 days. Expired medicines cannot be dispensed.
BR-08	Bed state machine. AVAILABLE→RESERVED, AVAILABLE→OCCUPIED, AVAILABLE→MAINTENANCE, RESERVED→OCCUPIED, RESERVED→AVAILABLE, OCCUPIED→CLEANING, CLEANING→AVAILABLE, CLEANING→MAINTENANCE, MAINTENANCE→AVAILABLE.
BR-09	Admission life cycle. ADMITTED → DISCHARGE_PENDING → DISCHARGED. Bed days = ceil(hours stayed / 24), minimum 1, charged at the ward's daily rate per stay.
BR-10	Invoice life cycle. UNPAID → PARTIALLY_PAID → PAID. UNPAID → CANCELLED (only when no payment exists). Discount 0 to 50% only while UNPAID.
BR-11	Payments. Amount > 0 and ≤ outstanding balance. Methods CASH, CARD, UPI, receipt number R-YYYYMMDD-NNNN.
BR-12	Account lockout. 5 consecutive failures lock the account for 15 minutes. A successful login resets the counter.

5.2 Authentication & Access Control (AUTH)

Table 8. Functional requirements (Authentication & Access Control)

ID	Requirement	Priority
FR-01	The system shall authenticate users with a username and password and issue a session token valid for 8 hours.	High
FR-02	The system shall lock an account for 15 minutes after 5 consecutive failed login attempts and shall reject logins for deactivated accounts.	High
FR-03	The system shall restrict every operation according to the role-permission matrix (Administrator, Receptionist, Doctor, Lab Technician, Pharmacist, Billing Staff) and show each role only its permitted menus.	High
FR-04	The system shall allow a logged-in user to log out and shall reject requests carrying a missing, invalid or expired token.	Medium

5.3 User Administration (USR)

Table 9. Functional requirements (User Administration)

ID	Requirement	Priority
FR-05	The Administrator shall be able to create user accounts with a unique username, full name, role and initial password (minimum 8 characters).	High
FR-06	The Administrator shall be able to activate/deactivate user accounts and reset passwords. An Administrator cannot deactivate their own account.	Medium

5.4 Patient Management (PAT)

Table 10. Functional requirements (Patient Management)

ID	Requirement	Priority
FR-07	The Receptionist shall be able to register a patient. The system shall generate a unique patient ID in the format P00001.	High
FR-08	The system shall validate patient data. First and last name required, 10-digit mobile number, date of birth not in the future, valid gender and blood group values.	High

ID	Requirement	Priority
FR-09	The system shall warn about a possible duplicate registration when a patient with the same mobile number and date of birth, or the same full name and date of birth, already exists, the Receptionist may confirm and proceed.	Medium
FR-10	Authorised users shall be able to search patients by patient ID, name or mobile number.	High
FR-11	The Receptionist shall be able to update patient demographic and contact details. The patient ID shall never change.	Medium
FR-12	The system shall display a patient's history, namely appointments, consultations, prescriptions, lab results, admissions and invoices (clinical items only to clinical roles).	High

5.5 Doctor & Department Management (DOC)

Table 11. Functional requirements (Doctor & Department Management)

ID	Requirement	Priority
FR-13	The Administrator shall be able to manage doctor profiles (department, specialisation, qualification, consultation fee, working days and hours, slot length and average consultation time).	High
FR-14	The system shall list departments and the doctors belonging to each department.	Low

5.6 Appointment Management (APT)

Table 12. Functional requirements (Appointment Management)

ID	Requirement	Priority
FR-15	The system shall show the free appointment slots of a doctor for a selected date, based on working days, working hours, slot length and existing bookings.	High
FR-16	The Receptionist shall be able to book an appointment for a registered patient with a doctor in a free slot that is not in the past and lies within the doctor's working hours.	High
FR-17	The system shall reject a booking that overlaps an existing active appointment of the same doctor or of the same patient.	High
FR-18	The Receptionist shall be able to reschedule an appointment while it is in the SCHEDULED state, subject to the same slot rules.	Medium
FR-19	The Receptionist shall be able to cancel a SCHEDULED appointment with a mandatory reason. The freed slot becomes bookable again.	Medium
FR-20	The system shall list appointments filtered by date, doctor and status.	Medium

5.7 Smart Queue Management (QUE)

Table 13. Functional requirements (Smart Queue Management)

ID	Requirement	Priority
FR-21	On check-in of a SCHEDULED appointment for today, the system shall issue a queue token unique per doctor per day and set the appointment state to CHECKED_IN.	High
FR-22	The system shall display a live queue board per doctor showing waiting, in-consultation and served patients, ordered by priority (URGENT first) and then check-in time.	High
FR-23	The system shall estimate each waiting patient's waiting time as (patients ahead × doctor's average consultation time) + remaining time of the consultation in progress.	High
FR-24	Receptionists, Doctors and Administrators shall be able to mark a waiting patient URGENT with a mandatory reason. The action shall be audited.	Medium
FR-25	A Doctor shall be able to call the next patient from their queue only when no consultation of theirs is in progress. The appointment moves to IN_CONSULTATION.	High

5.8 Consultation & Medical Records (CON)

Table 14. Functional requirements (Consultation & Medical Records)

ID	Requirement	Priority
FR-26	A Doctor shall be able to record a consultation for an appointment in IN_CONSULTATION (symptoms, vitals, diagnosis, clinical notes and optional follow-up date). Diagnosis is required to complete it.	High
FR-27	A Doctor shall be able to issue a prescription of one or more medicines with dosage, frequency, duration and quantity (> 0) during a consultation.	High
FR-28	A Doctor shall be able to order one or more lab tests for a patient during a consultation or an admission.	High
FR-29	Completing a consultation shall set the appointment to COMPLETED and add the doctor's consultation fee as an unbilled charge.	High
FR-30	Only clinical roles (Doctor, Administrator) shall view diagnoses and clinical notes. Receptionist and Billing Staff shall be denied.	High

5.9 Laboratory Management (LAB)

Table 15. Functional requirements (Laboratory Management)

ID	Requirement	Priority
FR-31	The Lab Technician shall see a worklist of lab orders filtered by status (ORDERED, SAMPLE_COLLECTED, COMPLETED, CANCELLED).	High
FR-32	The Lab Technician shall record sample collection (ORDERED → SAMPLE_COLLECTED) and then enter the result (SAMPLE_COLLECTED → COMPLETED). Other transitions shall be rejected.	High
FR-33	The system shall automatically flag a numeric result outside the test's reference range as ABNORMAL.	Medium
FR-34	Ordering a lab test shall add the test price as an unbilled charge. Cancelling an ORDERED test shall remove the unbilled charge.	Medium

5.10 Pharmacy & Inventory (PHA)

Table 16. Functional requirements (Pharmacy & Inventory)

ID	Requirement	Priority
FR-35	The Pharmacist shall manage the medicine inventory (add medicines, restock quantities and record batch and expiry date).	High
FR-36	The system shall raise alerts for medicines at or below their reorder level and for medicines expiring within 30 days or already expired.	Medium
FR-37	The Pharmacist shall dispense a PENDING prescription only if every item is in stock and not expired. Dispensing reduces stock and adds medicine charges.	High

5.11 Admission, Bed & Discharge Management (IPD)

Table 17. Functional requirements (Admission, Bed & Discharge Management)

ID	Requirement	Priority
FR-38	The system shall show a ward-wise bed board with each bed's status. AVAILABLE, RESERVED, OCCUPIED, CLEANING or MAINTENANCE.	High
FR-39	The Receptionist shall admit a patient to an AVAILABLE bed. A patient may have only one active admission and a bed may hold only one patient.	High
FR-40	The Receptionist shall transfer an admitted patient to another AVAILABLE bed. The previous bed moves to CLEANING and bed charges are kept per stay.	Medium

ID	Requirement	Priority
FR-41	Bed status changes shall follow the bed state machine (AVAILABLE→RESERVED→OCCUPIED→CLEANING→AVAILABLE, AVAILABLE ↔ MAINTENANCE, RESERVED→AVAILABLE), invalid transitions shall be rejected.	High
FR-42	A Doctor shall request discharge with a discharge summary (ADMITTED → DISCHARGE_PENDING). Bed charges (days × daily rate, minimum 1 day) are then added as unbilled charges.	High
FR-43	Discharge shall be completed only when all charges of the admission are invoiced and fully paid, unless an Administrator overrides with a recorded reason. The bed then moves to CLEANING.	High

5.12 Billing & Payments (BIL)

Table 18. Functional requirements (Billing & Payments)

ID	Requirement	Priority
FR-44	Billing Staff shall generate an invoice for a patient from all of that patient's unbilled charges (consultation, lab, medicine, bed), with a unique invoice number.	High
FR-45	Billing Staff shall be able to apply a discount of 0 to 50% with a reason to an UNPAID invoice.	Medium
FR-46	Billing Staff shall record full or partial payments (Cash, Card, UPI), a payment must be greater than zero and not exceed the outstanding balance. Invoice status moves UNPAID → PARTIALLY_PAID → PAID.	High
FR-47	The system shall generate a printable receipt with a unique receipt number for every payment and a printable invoice.	Medium

5.13 Dashboard & Reports (RPT)

Table 19. Functional requirements (Dashboard & Reports)

ID	Requirement	Priority
FR-48	The dashboard shall show today's KPIs. Appointments, patients waiting, bed occupancy, revenue collected, pending lab orders and stock alerts.	Medium
FR-49	The Administrator shall view reports for a date range (appointments per day and status, revenue per day and payment method, bed occupancy per ward and top diagnoses).	Medium

5.14 Audit Trail (AUD)

Table 20. Functional requirements (Audit Trail)

ID	Requirement	Priority
FR-50	The system shall record an append-only audit entry (user, role, action, entity, entity id, timestamp, summary) for every create, update and state change, without storing passwords. Administrators can search it.	High

6. Non-functional Requirements

Table 21. Non-functional requirements

ID	Category	Requirement
NFR-01	Performance	95% of API requests shall complete within 500 ms on a standard laptop with the demo dataset. Pages shall render within 2 seconds.
NFR-02	Security	Passwords shall be stored only as bcrypt hashes. All API routes except login shall require a signed JWT. All SQL shall use parameterised statements.
NFR-03	Security / Privacy	Access to clinical data shall follow the principle of least privilege defined in the role-permission matrix. Audit logs shall not contain passwords or tokens.
NFR-04	Reliability / Integrity	Multi-step operations (admission + bed update, dispensing + stock update, invoice generation, payment recording) shall execute in a single database transaction.

ID	Category	Requirement
NFR-05	Usability	The interface shall be responsive (≥ 360 px width), use consistent colour-coded status badges, show field-level validation messages and allow booking an appointment in at most 4 steps.
NFR-06	Maintainability	The code shall follow a layered architecture (UI → REST API → Services → Domain rules → Data access). Business rules shall be isolated in pure, unit-tested domain modules.
NFR-07	Portability	The system shall run on Windows, macOS and Linux with Node.js 20 or later, and on current Chrome, Edge, Firefox and Safari browsers.
NFR-08	Auditability	Every state-changing operation shall be traceable to an authenticated user and timestamp through the audit trail.
NFR-09	Availability	The production deployment should target 99.5% availability during hospital operating hours, with daily database backups (out of scope for the prototype).
NFR-10	Scalability	The design shall support at least 50 concurrent staff users and 100,000 patient records without architectural change.

NFR-01 to NFR-08 are verified by automated tests (see the Software Test Report). NFR-09 and NFR-10 are design targets for a production deployment and are addressed by the architecture (see the Software Design Document) rather than measured on the prototype.

7. User Requirements

User requirements describe, in the language of each user class, what they need to achieve. They are written as user stories with acceptance criteria and are refined into the functional requirements listed in the last column. The acceptance tests (AT-01 … AT-05) in the Software Test Report validate them end-to-end.

Table 22. User requirements (user stories)

ID	As a …	I want to …	So that …	Acceptance criteria	FRs
UR-01	Receptionist	register a new patient in under two minutes and be warned if the patient may already exist	so that every patient has exactly one record and one permanent ID	Unique ID P00001… is shown on save. Invalid mobile/DOB is highlighted on the field. A possible duplicate (same mobile + DOB or same name + DOB) is listed and must be confirmed.	FR-07, FR-08, FR-09
UR-02	Receptionist / Doctor	find a patient by ID, name or mobile and open the full history	so that no information has to be asked or typed twice	Search returns matches as I type. The history shows visits, admissions and bills, clinical notes are visible to doctors only.	FR-10, FR-11, FR-12, FR-30
UR-03	Receptionist	book, reschedule and cancel appointments only in free slots	so that doctors are never double-booked	Free / booked / past slots are shown for the chosen date. Booking a taken slot is refused with a clear message. Cancellation asks for a reason.	FR-15 to FR-20
UR-04	Receptionist	check patients in and give them a token with an expected waiting time	so that the waiting area is orderly and patients know when they will be seen	Token numbers are sequential per doctor per day. The queue board shows position and ETA. Urgent patients can be moved to the front with a recorded reason.	FR-21 to FR-24
UR-05	Doctor	call the next patient and record vitals, diagnosis, prescription and lab orders on one screen	so that the consultation is fast and complete	Only one consultation can be open at a time. Diagnosis is mandatory to complete. Completing the visit charges the fee automatically.	FR-25 to FR-29
UR-06	Lab Technician	see a worklist of pending tests and enter results that are flagged when abnormal	so that urgent abnormal results reach the doctor quickly	STAT orders appear first, a result can only be entered after sample collection. Values outside the reference range are marked ABNORMAL.	FR-31 to FR-34

ID	As a ...	I want to ...	So that ...	Acceptance criteria	FRs
UR-07	Pharmacist	dispense prescriptions only when stock is sufficient and be alerted about low or expiring stock	so that patients never receive expired medicine and stock-outs are avoided	Dispensing is blocked when any item is short or expired. Low-stock and expiry (≤ 30 days) alerts are listed. Restocking records batch and expiry.	FR-35 to FR-37
UR-08	Receptionist	see which beds are free and admit or transfer patients safely	so that two patients are never assigned the same bed	The bed board is colour-coded by status. Only AVAILABLE/RESERVED beds can be assigned. A transferred patient's old bed goes to CLEANING.	FR-38 to FR-41
UR-09	Doctor / Receptionist	discharge a patient only after the bill is settled, with an emergency override for the administrator	so that revenue is not lost and exceptions are accountable	Discharge is refused while charges are uninvoiced or unpaid. An administrator can override with a mandatory reason that is audited.	FR-42, FR-43
UR-10	Billing Staff	generate an invoice from all pending charges, apply approved discounts and accept part payments	so that billing is accurate and quick	Invoice lines are filled automatically from consultations, tests, medicines and bed days, discount 0 to 50%, overpayment is refused, each payment prints a receipt.	FR-44 to FR-47
UR-11	Administrator	manage staff accounts, roles and doctor schedules	so that each person can do exactly their job	Usernames are unique, passwords ≥ 8 characters, accounts can be (de)activated, doctor working days, hours and fees are editable.	FR-05, FR-06, FR-13, FR-14
UR-12	Administrator	see today's key figures and period reports	so that I can monitor workload, occupancy and revenue	Dashboard KPIs per role, reports for a date range with charts and CSV export.	FR-48, FR-49
UR-13	All staff	log in securely and see only the functions of my role	so that patient data is protected	Account locks for 15 minutes after 5 wrong passwords. Menus and APIs are restricted by role. I can log out.	FR-01 to FR-04
UR-14	Administrator	trace who did what and when	so that clinical and financial actions are accountable	Every state change is in an append-only audit trail that can be searched by entity, action, user and date.	FR-50

8. System Requirements

System requirements refine the user requirements into precise behaviour of the software system, its interfaces and its data.

8.1 System features (stimulus → response)

Table 23. System features

Feature	Stimulus / input	System response / output
Authentication	Username + password. Bearer token on every API call	Issues an 8-hour JWT bound to a server-side session. Increments failed attempts and locks after 5. Rejects revoked/expired tokens with HTTP 401 and unauthorised roles with HTTP 403.
Patient registration	Demographics, contact, blood group, allergies	Validates fields, searches existing records with the same DOB for phone/name matches. Returns HTTP 409 POSSIBLE_DUPLICATE unless confirmed. Assigns the next ID P#####.
Slot calculation	Doctor + date	Generates slots from work start to end at slot length on working days. Marks slots BOOKED (overlap with an active appointment) or PAST (before current time).
Appointment booking	Patient, doctor, date, start time, reason	Validates working day/hours, slot alignment and future time. Rejects doctor or patient overlap (HTTP 409). Stores SCHEDULED appointment in a transaction.

Feature	Stimulus / input	System response / output
Check-in & queue	Today's SCHEDULED appointment	Assigns token = max(token)+1 for doctor/day. Sets CHECKED_IN, queue ordered URGENT → check-in time. ETA = ahead × average + remaining time of current consultation.
Consultation	Vitals, symptoms, diagnosis, notes, prescription items, lab tests	Validates vitals ranges and BP format. Stores prescription (PENDING) and lab orders (ORDERED) with charges. On completion requires diagnosis, closes appointment and adds the consultation fee charge.
Laboratory	Sample collected, result value / findings	Enforces ORDERED → SAMPLE_COLLECTED → COMPLETED. Computes ABNORMAL when the value lies outside [ref_low, ref_high], cancelling removes the unbilled charge.
Pharmacy	Restock (qty, batch, expiry), dispense request	Refuses expired batches, checks every item for stock and expiry inside one transaction. Decrements stock and creates medicine charges. Produces low-stock / near-expiry / expired alerts.
Beds & admissions	Patient, attending doctor, bed, transfer bed, discharge	Allows only valid bed transitions. One active admission per patient and one patient per bed (enforced by unique indexes). Bed-day charges = max(1, ⌈hours/24⌉) shared across stays, discharge only when clearance holds or an administrator overrides with a reason.
Billing	Patient (+ admission), discount %, payment amount, method, reference	Builds an invoice from unbilled charges with number INV-YYYYMMDD-NNNN. Discount 0 to 50% only while UNPAID. Payment 0 < amount ≤ balance. Reference required for CARD/UPI, receipt R-YYYYMMDD-NNNN.
Reporting	Date range	Aggregates appointments by day/status, revenue by day/method/source, occupancy by ward, top diagnoses and doctor performance.
Audit	Any create / update / state change	Appends user, role, action, entity, id, timestamp and summary, credentials are scrubbed. UPDATE/DELETE on the log are blocked by database triggers.

8.2 External interface requirements

User interface. A responsive single-page web application (minimum width 360 px) with a sidebar showing only the modules permitted for the logged-in role, colour-coded status badges, modal dialogs for actions, field-level validation messages, toast notifications and print-optimised invoice and receipt layouts.

Software interface (REST API). All functions are exposed as JSON over HTTP under `/api`. Every route except login and health requires a Bearer token. Each is protected by the permission shown below (generated from the source code).

Table 24. REST API endpoints (78)

Method	Endpoint	Required permission
GET	`/api/health`	public
POST	`/api/auth/login`	public
GET	`/api/auth/me`	any authenticated user
POST	`/api/auth/logout`	any authenticated user
GET	`/api/users`	users.manage
POST	`/api/users`	users.manage
PATCH	`/api/users/{id}`	users.manage
POST	`/api/users/{id}/activate`	users.manage
POST	`/api/users/{id}/deactivate`	users.manage
POST	`/api/users/{id}/reset-password`	users.manage
GET	`/api/departments`	doctors.read
POST	`/api/departments`	doctors.manage

Method	Endpoint	Required permission
GET	/api/doctors	doctors.read
GET	/api/doctors/{id}	doctors.read
POST	/api/doctors	doctors.manage
PATCH	/api/doctors/{id}	doctors.manage
GET	/api/doctors/{id}/slots	appointments.read
GET	/api/patients	patients.read
POST	/api/patients/check-duplicates	patients.write
POST	/api/patients	patients.write
GET	/api/patients/{id}	patients.read
PATCH	/api/patients/{id}	patients.write
GET	/api/patients/{id}/history	patients.read
GET	/api/appointments	appointments.read
POST	/api/appointments	appointments.write
GET	/api/appointments/{id}	appointments.read
PATCH	/api/appointments/{id}/reschedule	appointments.write
POST	/api/appointments/{id}/cancel	appointments.write
POST	/api/appointments/{id}/no-show	appointments.write
POST	/api/appointments/{id}/check-in	queue.checkin
GET	/api/queue	queue.read
GET	/api/queue/{doctorId}	queue.read
POST	/api/queue/{appointmentId}/priority	queue.priority
POST	/api/queue/{doctorId}/call-next	consultation.write
GET	/api/consultations	clinical.read
GET	/api/consultations/current	consultation.write
GET	/api/consultations/{id}	clinical.read
PUT	/api/consultations/{id}	consultation.write
PUT	/api/consultations/{id}/prescription	consultation.write
POST	/api/consultations/{id}/complete	consultation.write
GET	/api/lab/tests	lab.read or lab.order
GET	/api/lab/orders	lab.read
GET	/api/lab/stats	lab.read
POST	/api/lab/orders	lab.order
GET	/api/lab/orders/{id}	lab.read
POST	/api/lab/orders/{id}/collect	lab.process
POST	/api/lab/orders/{id}/result	lab.process
POST	/api/lab/orders/{id}/cancel	lab.process or lab.order
GET	/api/pharmacy/medicines	pharmacy.read
POST	/api/pharmacy/medicines	pharmacy.manage
PATCH	/api/pharmacy/medicines/{id}	pharmacy.manage
POST	/api/pharmacy/medicines/{id}/restock	pharmacy.manage
GET	/api/pharmacy/alerts	pharmacy.read
GET	/api/pharmacy/prescriptions	prescriptions.read
GET	/api/pharmacy/prescriptions/{id}	prescriptions.read

Method	Endpoint	Required permission
POST	`/api/pharmacy/prescriptions/{id}/dispense`	prescriptions.dispense
GET	`/api/beds`	beds.read or admissions.read
GET	`/api/beds/available`	admissions.write
POST	`/api/beds/{id}/status`	beds.manage
GET	`/api/admissions`	admissions.read
POST	`/api/admissions`	admissions.write
GET	`/api/admissions/{id}`	admissions.read
POST	`/api/admissions/{id}/transfer`	admissions.write
POST	`/api/admissions/{id}/request-discharge`	admissions.request-discharge
POST	`/api/admissions/{id}/discharge`	admissions.write
GET	`/api/billing/pending`	billing.read
GET	`/api/billing/unbilled`	billing.read
GET	`/api/billing/invoices`	billing.read
POST	`/api/billing/invoices`	billing.write
GET	`/api/billing/invoices/{id}`	billing.read
POST	`/api/billing/invoices/{id}/discount`	billing.write
POST	`/api/billing/invoices/{id}/payments`	billing.write
POST	`/api/billing/invoices/{id}/cancel`	billing.write
GET	`/api/billing/receipts/{paymentId}`	billing.read
GET	`/api/reports/dashboard`	dashboard.read
GET	`/api/reports/summary`	reports.read
GET	`/api/audit`	audit.read
GET	`/api/audit/entities`	audit.read

Hardware interfaces. Standard keyboard/mouse or touch screen. Any printer supported by the browser for invoices and receipts. **Communication interfaces.** HTTP(S) on a hospital LAN. TLS termination by a reverse proxy in production.

8.3 Data requirements

- Persistent data is stored in a relational database (SQLite file) with 20 tables (users, sessions, departments, doctors, patients, appointments, consultations, prescriptions, prescription items, medicines, lab tests, lab orders, wards, beds, bed stays, admissions, charges, invoices, payments and audit logs).
- Integrity rules are enforced in the schema, namely foreign keys, CHECK constraints on every status column, a unique token per doctor per day, at most one active admission per patient and one admission per bed (partial unique indexes), and append-only triggers on the audit log.
- Money is stored in rupees with two decimals and calculated in paise to avoid rounding drift. Dates are ISO-8601, appointment dates/times are local (IST).
- Medical and billing records are never deleted. Cancellations change status and are audited.

9. Hardware and Software Requirements

9.1 Hardware requirements

Table 25. Hardware requirements

Component	Minimum	Recommended
Application server	Dual-core 2 GHz CPU, 4 GB RAM, 2 GB free disk	Quad-core CPU, 8 GB RAM, SSD with 20 GB free, UPS
Client workstation	Any PC, laptop or tablet able to run a current browser, 1366 × 768 display	1920 × 1080 display at reception and nursing stations. Tablet for doctors
Network	100 Mbit/s LAN or Wi-Fi	Gigabit LAN, separate VLAN for clinical systems
Peripherals	Any printer for invoices/receipts	Thermal receipt printer at billing. Barcode scanner (future)

9.2 Software requirements

Table 26. Software requirements

Layer	Software (version used)	Purpose
Operating system	Windows 10/11, macOS 12+, Ubuntu 22.04+	Server and development machines
Runtime	Node.js 20 or later (tested on v24.21.0), npm 10	Runs the API server and build tools
Web server / API	Express 4.22.3, jsonwebtoken 9.0.3, bcryptjs 2.4.3, zod 3.25.76	REST API, authentication, password hashing, input validation
Database	SQLite 3 via better-sqlite3 12.11.1	Embedded relational database (single file, WAL mode)
Front end	React 18.3.1, React Router 6.30.6, Tailwind CSS 3.4.19, Recharts 2.15.4, Vite 5.4.21	Single-page user interface and build
Browser	Chrome / Edge 120+, Firefox 120+, Safari 16+	Client access
Testing	Vitest 3.2.4, Supertest 7.3.1, Playwright 1.64.1	Unit, integration, system and acceptance tests
Design & documentation	PlantUML, docx (document generator), LibreOffice	UML diagrams and this report
Version control	Git, GitHub	Source control and collaboration

10. Assumptions and Constraints

10.1 Assumptions

- The system serves a single hospital site. All staff have individual accounts created by the administrator.
- Patients do not access the system directly. The receptionist registers them and books appointments.
- Doctors work a fixed weekly schedule (days and hours) with a constant slot length. Leave management is out of scope.
- One consultation is recorded per appointment. In-patient lab tests are ordered against the admission.
- Consultation and in-patient services are treated as tax-exempt, so no GST is computed. All amounts are in Indian rupees (₹).
- Payments are taken at the hospital counter (cash, card or UPI) and only recorded in MediFlow. No money is processed online.
- The server clock is correct and set to the hospital time zone (IST).
- All demonstration data (patients, doctors, bills) is synthetic, no real patient data is used.

10.2 Constraints

- **Schedule.** The project had to be specified, designed, implemented and tested by a group of five students before 31 October 2026.
- **Technology.** Only free and open-source software is used. The system must run on a standard student laptop without a separate database server.
- **Platform.** Browser-based only, no native mobile app.
- **Regulatory.** Patient data is sensitive. Access must follow least privilege, passwords must be hashed and every change must be auditable. Full compliance with the Digital Personal Data Protection Act 2023 and NABH accreditation standards would require further work (consent management, data retention policy, backups) beyond this prototype.
- **Scalability.** SQLite supports one application server. A multi-branch deployment would move to a client to server DBMS (e.g. PostgreSQL) without changing the service layer.

10.3 Dependencies

- Availability of npm packages listed in Section 9.2 at build time.
- A modern browser with JavaScript enabled on every client.